OS ASSIGNMENT

PHASE 1 — object.c

> - Integrity checking (detects corrupted objects)
>
> 📸 **Screenshot 1A:** Output of `./test_objects` showing all tests passing.
>
> 📸 **Screenshot 1B:** `find .pes/objects -type f` showing the sharded directory structure.

ANS 1A

```
┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ make test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto

┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B

```
┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
```

PHASE 2 — tree.c

> 📸 **Screenshot 2A:** Output of `./test_tree` showing all tests passing.
>
> 📸 **Screenshot 2B:** Pick a tree object from `find .pes/objects -type f` and run `xxd .pes/objects/XX/YYY... | head -20` to show the raw binary format.

2A

```
┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B

```
┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
```

Phase 3: The Index

3A

```
┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ make pes
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto

┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ ./pes init
Initialized empty PES repository in .pes/

┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ echo "hello" > file1.txt
echo "world" > file2.txt

┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ ./pes add file1.txt file2.txt

┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  object.c
  untracked:  test.txt
  untracked:  pes.c
  untracked:  index.h
  untracked:  commit.c
  untracked:  pes.h
  untracked:  test_objects
  untracked:  some_file.txt
  untracked:  commit.h
  untracked:  Makefile
  untracked:  README.md
  untracked:  tree.h
  untracked:  tree.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  .gitignore
  untracked:  test_tree
```

```
  untracked:   some_file.txt
  untracked:   commit.h
  untracked:   Makefile
  untracked:   README.md
  untracked:   tree.h
  untracked:   tree.c
  untracked:   test_tree.c
  untracked:   test_objects.c
  untracked:   .gitignore
  untracked:   test_tree
  untracked:   test_sequence.sh
  untracked:   index.c


┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ |
```

3B

```
┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776413610 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776413610 6 file2.txt
```

4A

```
┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ ./pes log
commit 3a2a6469383a6b0b7995b604c3f62e55cf7cf9810f4d842e7422a6fa6150cf27
Author: MADHU <PES2UG24CS210>
Date:   1776428763

    Add farewell


commit cf93ad1264cb487d5c99965b0d0c781d3667e647a09393e00550b0b4863c4dd4
Author: MADHU <PES2UG24CS210>
Date:   1776428748

    Add world


commit 84205a687d2fc1b4dc28d3b7347202492e411c48d5f28c8fa936d0dea29d8ea0
Author: MADHU <PES2UG24CS210>
Date:   1776428677

    Initial commit
```

4B

```
┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ find .pes -type f | sort
.pes/HEAD
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/3a/2a6469383a6b0b7995b604c3f62e55cf7cf9810f4d842e7422a6fa6150cf27
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/84/205a687d2fc1b4dc28d3b7347202492e411c48d5f28c8fa936d0dea29d8ea0
.pes/objects/cf/93ad1264cb487d5c99965b0d0c781d3667e647a09393e00550b0b4863c4dd4
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

4C

```
┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ cat .pes/refs/heads/main
3a2a6469383a6b0b7995b604c3f62e55cf7cf9810f4d842e7422a6fa6150cf27

┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ cat .pes/HEAD
ref: refs/heads/main

┌──(madhu㉿madhusasus)-[~/PES2UG24CS210-pes-vcs]
└─$ |
```

Phase 5 & 6: Analysis-Only Questions

Q5.1 — pes checkout <branch>: Update .pes/HEAD to point to new branch, read that branch's commit → tree, then update every file in working directory to match. Complexity: handling conflicts, deletions, new files.

Q5.2 — Dirty working directory detection: Compare working file's mtime/size against index entry. If different, stat-compare or rehash. Also compare index entry against HEAD commit's tree hash.

Q5.3 — Detached HEAD: Commits are made but no branch pointer advances. Commits become unreachable after moving HEAD. Recovery: git reflog shows recent HEAD values; create a branch pointing to the last known hash.

Q6.1 — GC algorithm: Start from all branch refs → walk commit chain → collect all tree/blob hashes reachable. Use a HashSet. Delete everything in .pes/objects/ not in the set. For 100K commits × ~5 objects each = ~500K objects to visit.

Q6.2 — Race condition: GC marks object as unreachable → concurrent commit writes new commit referencing that object → GC deletes it → repo corrupted. Git avoids this with a grace period (objects newer than 2 weeks are never deleted).